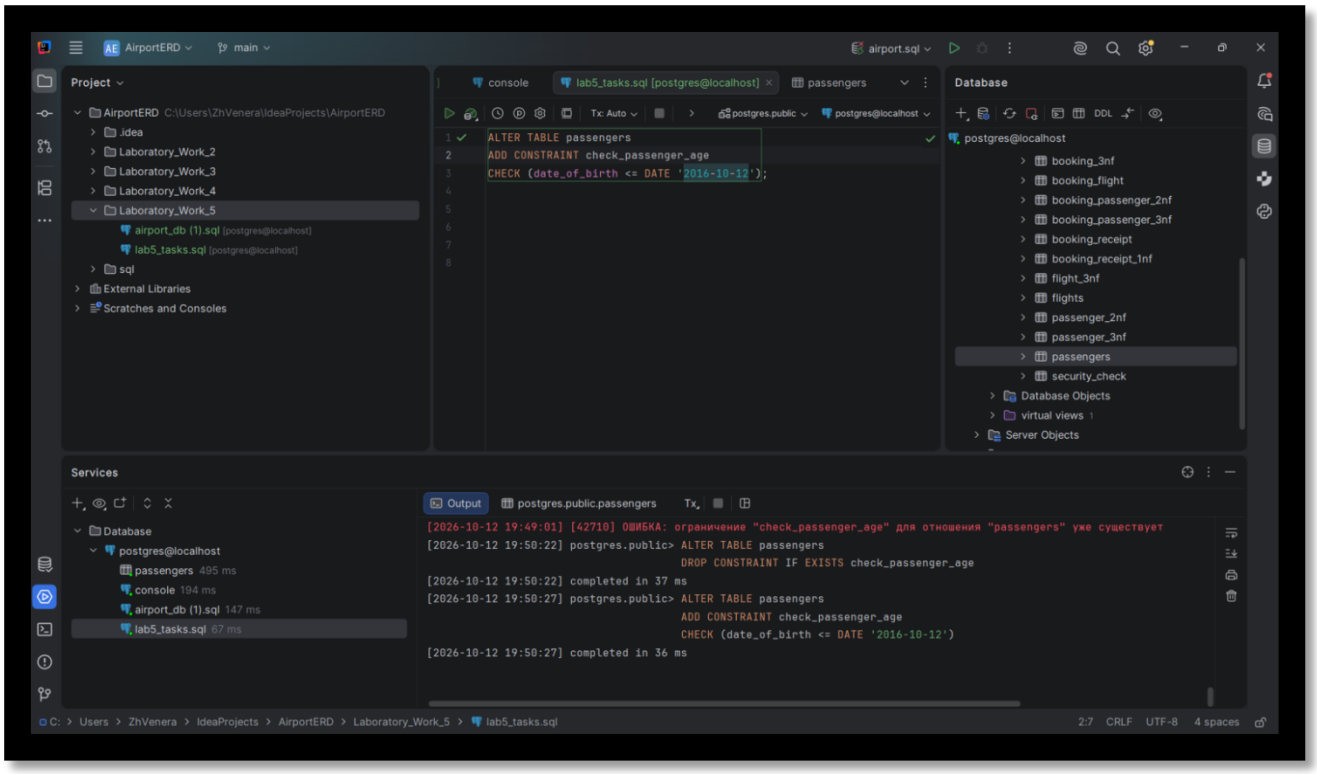
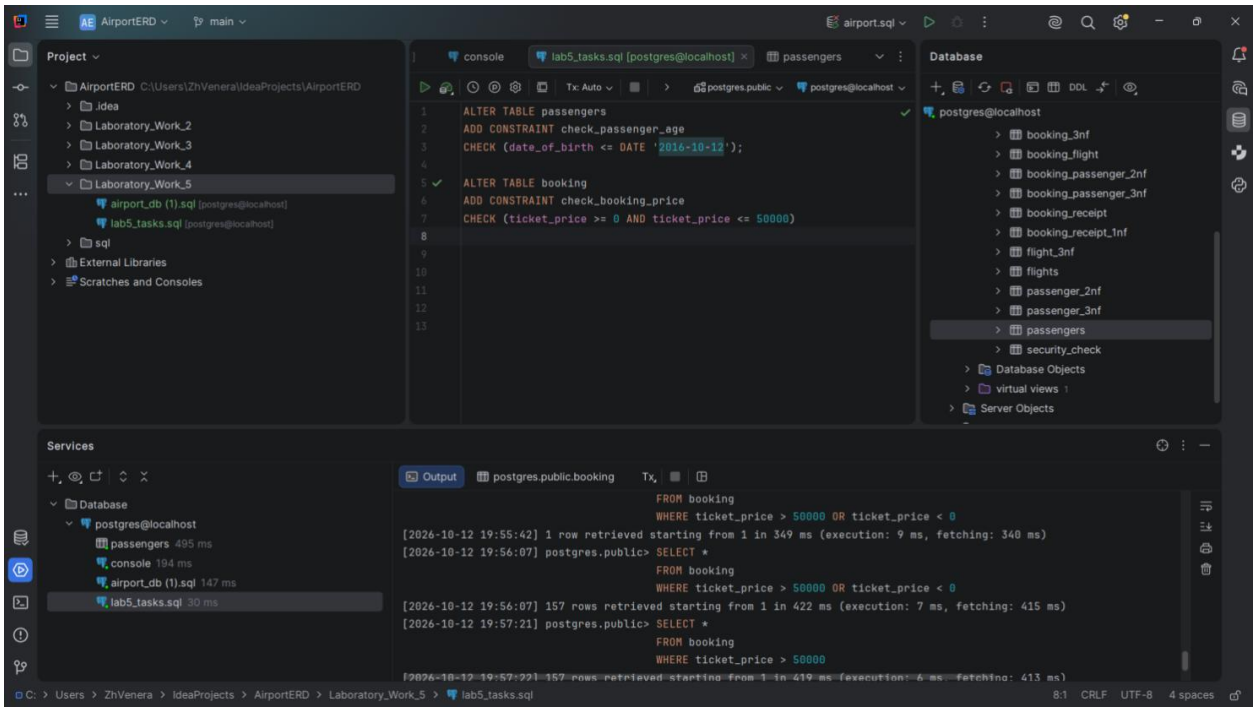
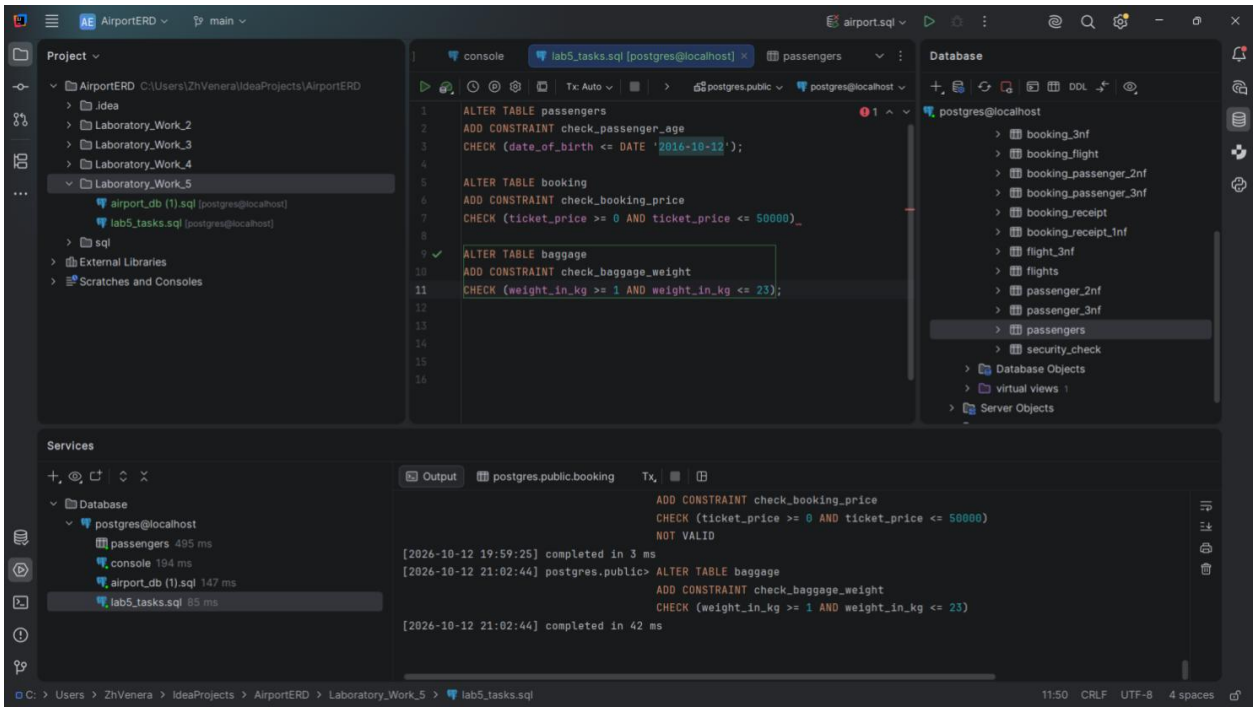
TASK 1


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql [postgres@localhost]
lab5_tasks.sql [postgres@localhost]
sql
External Libraries
Scratches and Consoles
console
lab5_tasks.sql [postgres@localhost]
passengers
Tx: Auto
postgres.public
postgres@localhost
1 ALTER TABLE passengers
2 ADD CONSTRAINT check_passenger_age
3 CHECK (date_of_birth <= DATE '2016-10-12');
Database
DDL
postgres@localhost
booking_3nf
booking_flight
booking_passenger_2nf
booking_passenger_3nf
booking_receipt
booking_receipt_1nf
flight_3nf
flights
passenger_2nf
passenger_3nf
passengers
security_check
Database Objects
virtual views 1
Server Objects
Services
Database
postgres@localhost
passengers 495 ms
console 194 ms
airport_db (1).sql 147 ms
lab5_tasks.sql 67 ms
Output
postgres.public.passengers
[2026-10-12 19:49:01] [42710] ОШИБКА: ограничение "check_passenger_age" для отношения "passengers" уже существует
[2026-10-12 19:50:22] postgres.public> ALTER TABLE passengers
DROP CONSTRAINT IF EXISTS check_passenger_age
[2026-10-12 19:50:22] completed in 37 ms
[2026-10-12 19:50:27] postgres.public> ALTER TABLE passengers
ADD CONSTRAINT check_passenger_age
CHECK (date_of_birth <= DATE '2016-10-12')
[2026-10-12 19:50:27] completed in 36 ms
C: Users ZhVenera IdeaProjects AirportERD Laboratory_Work_5 lab5_tasks.sql
2:7 CRLF UTF-8 4 spaces

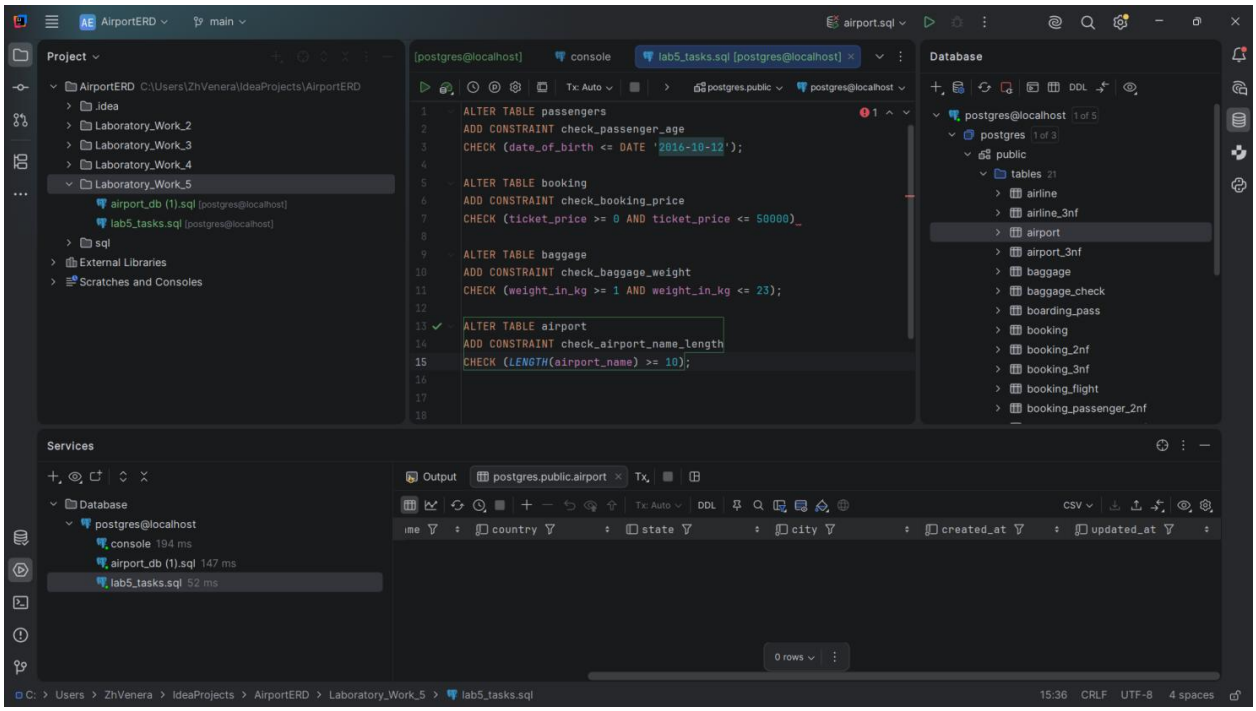
TASK 2


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql [postgres@localhost]
lab5_tasks.sql [postgres@localhost]
sql
External Libraries
Scratches and Consoles
console
lab5_tasks.sql [postgres@localhost]
passengers
Tx: Auto
postgres.public
postgres@localhost
1 ALTER TABLE passengers
2 ADD CONSTRAINT check_passenger_age
3 CHECK (date_of_birth <= DATE '2016-10-12');
5 ALTER TABLE booking
6 ADD CONSTRAINT check_booking_price
7 CHECK (ticket_price >= 0 AND ticket_price <= 50000)
Database
DDL
postgres@localhost
booking_3nf
booking_flight
booking_passenger_2nf
booking_passenger_3nf
booking_receipt
booking_receipt_1nf
flight_3nf
flights
passenger_2nf
passenger_3nf
passengers
security_check
Database Objects
virtual views 1
Server Objects
Services
Database
postgres@localhost
passengers 495 ms
console 194 ms
airport_db (1).sql 147 ms
lab5_tasks.sql 30 ms
Output
postgres.public.booking
FROM booking
WHERE ticket_price > 50000 OR ticket_price < 0
[2026-10-12 19:55:42] 1 row retrieved starting from 1 in 349 ms (execution: 9 ms, fetching: 340 ms)
[2026-10-12 19:56:07] postgres.public> SELECT *
FROM booking
WHERE ticket_price > 50000 OR ticket_price < 0
[2026-10-12 19:56:07] 157 rows retrieved starting from 1 in 422 ms (execution: 7 ms, fetching: 415 ms)
[2026-10-12 19:57:21] postgres.public> SELECT *
FROM booking
WHERE ticket_price > 50000
C: Users ZhVenera IdeaProjects AirportERD Laboratory_Work_5 lab5_tasks.sql
8:1 CRLF UTF-8 4 spaces

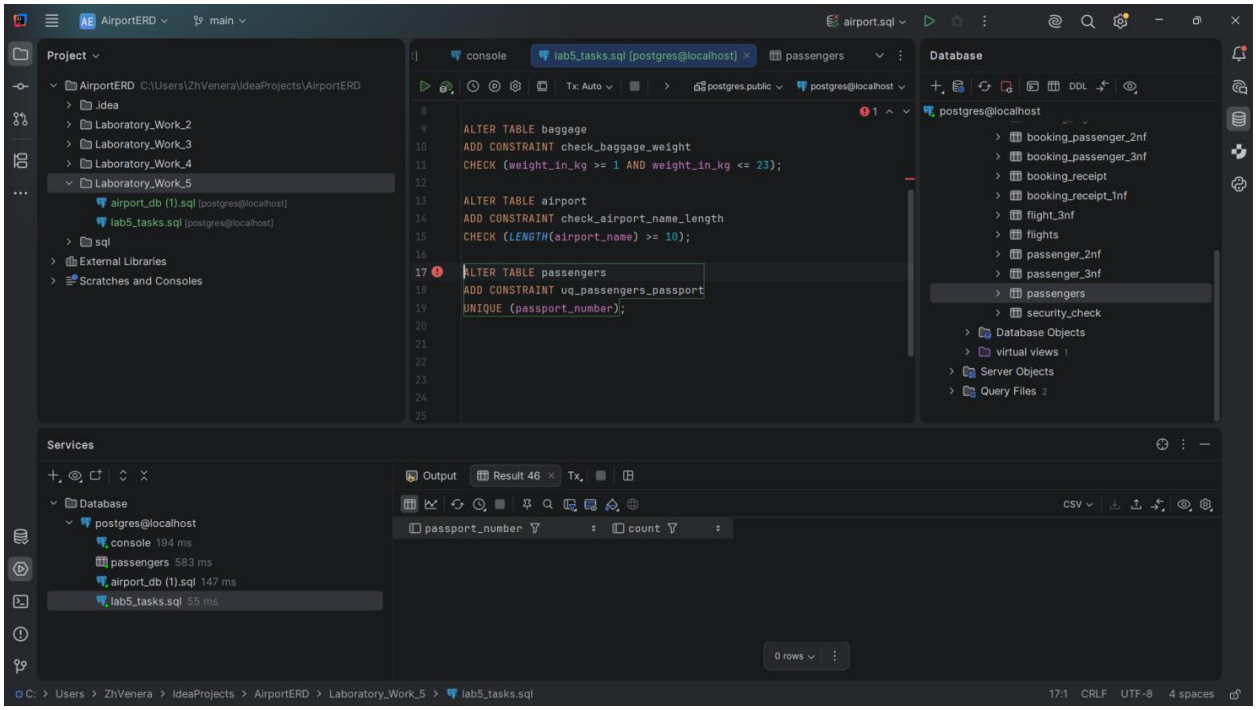
TASK 3


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql [postgres@localhost]
lab5_tasks.sql [postgres@localhost]
sql
External Libraries
Scratches and Consoles
console
lab5_tasks.sql [postgres@localhost]
passengers
Tx: Auto
postgres.public
postgres@localhost
ALTER TABLE passengers
ADD CONSTRAINT check_passenger_age
CHECK (date_of_birth <= DATE '2016-10-12');
ALTER TABLE booking
ADD CONSTRAINT check_booking_price
CHECK (ticket_price >= 0 AND ticket_price <= 50000)
ALTER TABLE baggage
ADD CONSTRAINT check_baggage_weight
CHECK (weight_in_kg >= 1 AND weight_in_kg <= 23);
Database
DDL
postgres@localhost
booking_3nf
booking_flight
booking_passenger_2nf
booking_passenger_3nf
booking_receipt
booking_receipt_1nf
flight_3nf
flights
passenger_2nf
passenger_3nf
passengers
security_check
Database Objects
virtual views
Server Objects
Services
Output
postgres.public.booking
Database
postgres@localhost
passengers 495 ms
console 194 ms
airport_db (1).sql 147 ms
lab5_tasks.sql 85 ms
ADD CONSTRAINT check_booking_price
CHECK (ticket_price >= 0 AND ticket_price <= 50000)
NOT VALID
[2026-10-12 19:59:25] completed in 3 ms
[2026-10-12 21:02:44] postgres.public> ALTER TABLE baggage
ADD CONSTRAINT check_baggage_weight
CHECK (weight_in_kg >= 1 AND weight_in_kg <= 23)
[2026-10-12 21:02:44] completed in 42 ms
C: Users ZhVenera IdeaProjects AirportERD Laboratory_Work_5 lab5_tasks.sql
11:50 CRLF UTF-8 4 spaces

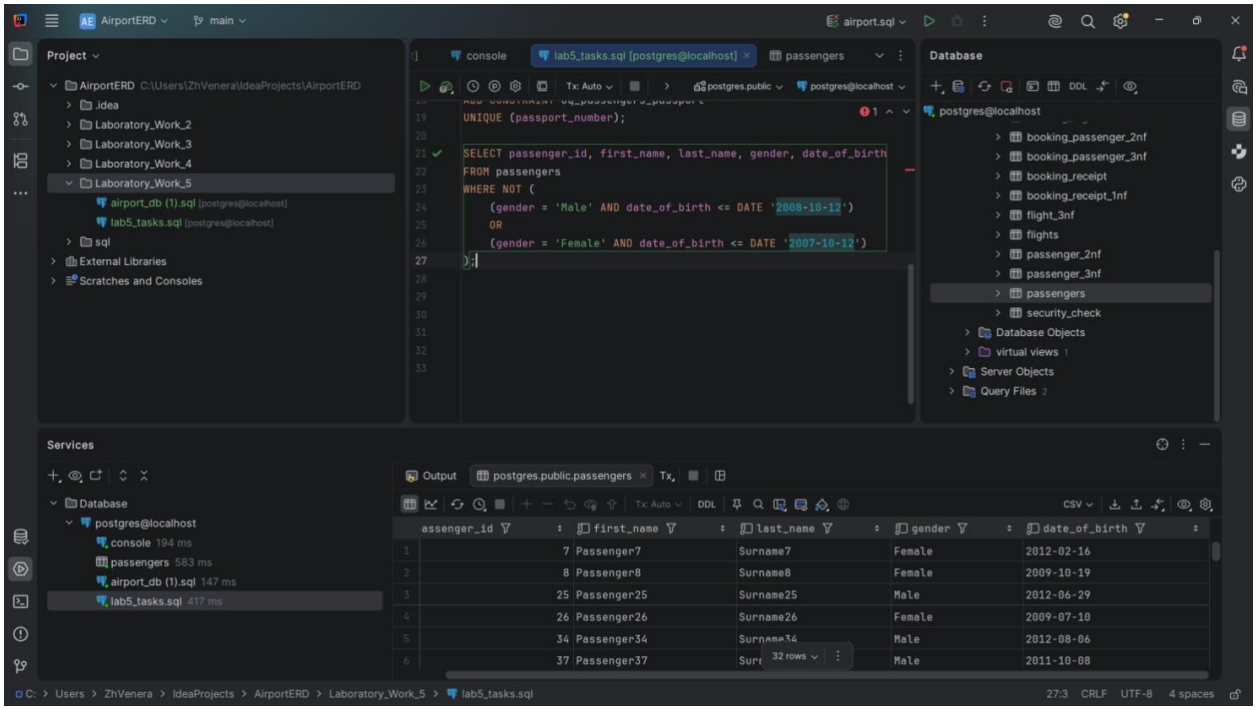
TASK 4


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql [postgres@localhost]
lab5_tasks.sql [postgres@localhost]
sql
External Libraries
Scratches and Consoles
[postgres@localhost]
console
lab5_tasks.sql [postgres@localhost]
Tx: Auto
postgres.public
postgres@localhost
ALTER TABLE passengers
ADD CONSTRAINT check_passenger_age
CHECK (date_of_birth <= DATE '2016-10-12');
ALTER TABLE booking
ADD CONSTRAINT check_booking_price
CHECK (ticket_price >= 0 AND ticket_price <= 50000)
ALTER TABLE baggage
ADD CONSTRAINT check_baggage_weight
CHECK (weight_in_kg >= 1 AND weight_in_kg <= 23);
ALTER TABLE airport
ADD CONSTRAINT check_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
Database
DDL
postgres@localhost 1 of 5
postgres 1 of 3
public
tables 21
airline
airline_3nf
airport
airport_3nf
baggage
baggage_check
boarding_pass
booking
booking_2nf
booking_3nf
booking_flight
booking_passenger_2nf
Services
Output
postgres.public.airport
Tx: Auto
DDL
CSV
country
state
city
created_at
updated_at
0 rows
Database
postgres@localhost
console 194 ms
airport_db (1).sql 147 ms
lab5_tasks.sql 52 ms
C: Users ZhVenera IdeaProjects AirportERD Laboratory_Work_5 lab5_tasks.sql
15:36 CRLF UTF-8 4 spaces

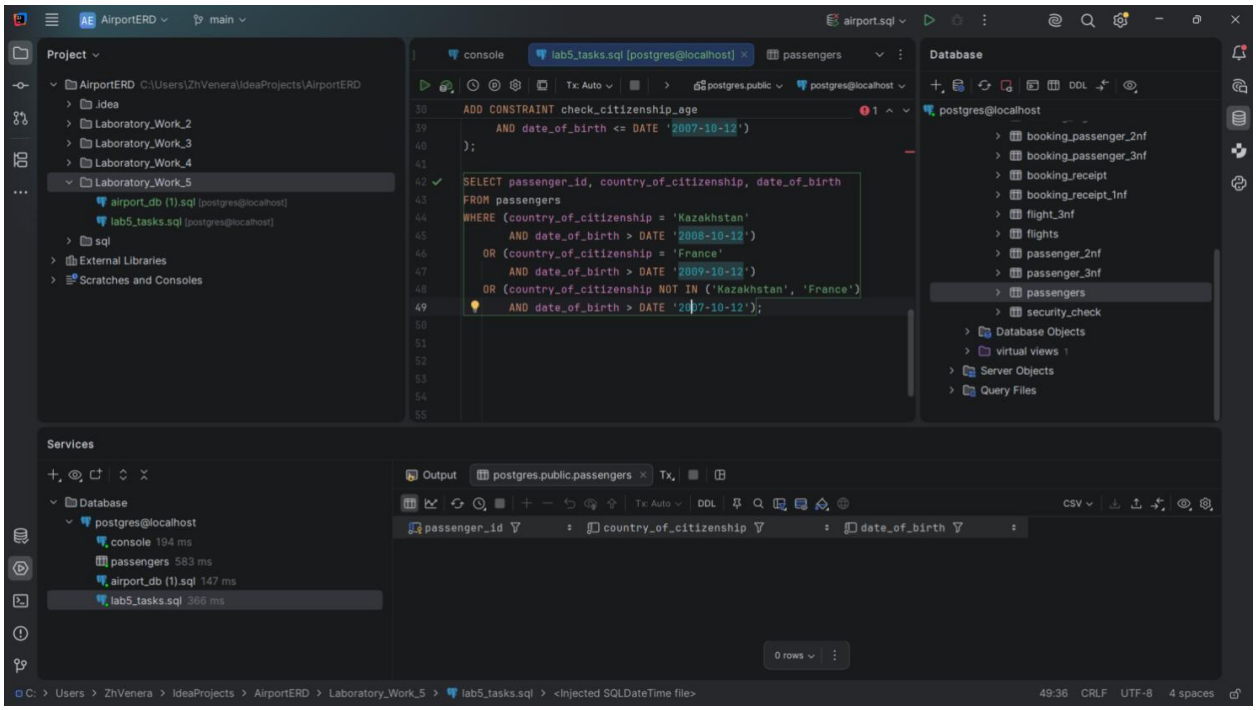
TASK 5


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql [postgres@localhost]
lab5_tasks.sql [postgres@localhost]
sql
External Libraries
Scratches and Consoles
console
lab5_tasks.sql [postgres@localhost]
passengers
Tx: Auto
postgres.public
postgres@localhost
ALTER TABLE baggage
ADD CONSTRAINT check_baggage_weight
CHECK (weight_in_kg >= 1 AND weight_in_kg <= 23);
ALTER TABLE airport
ADD CONSTRAINT check_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
ALTER TABLE passengers
ADD CONSTRAINT uq_passengers_passport
UNIQUE (passport_number);
Database
DDL
postgres@localhost
booking_passenger_2nf
booking_passenger_3nf
booking_receipt
booking_receipt_1nf
flight_3nf
flights
passenger_2nf
passenger_3nf
passengers
security_check
Database Objects
virtual views
Server Objects
Query Files
Services
Output
Result 46
Tx
CSV
Database
postgres@localhost
console 194 ms
passengers 583 ms
airport_db (1).sql 147 ms
lab5_tasks.sql 55 ms
passport_number
count
0 rows
C: Users ZhVenera IdeaProjects AirportERD Laboratory_Work_5 lab5_tasks.sql
17:1 CRLF UTF-8 4 spaces

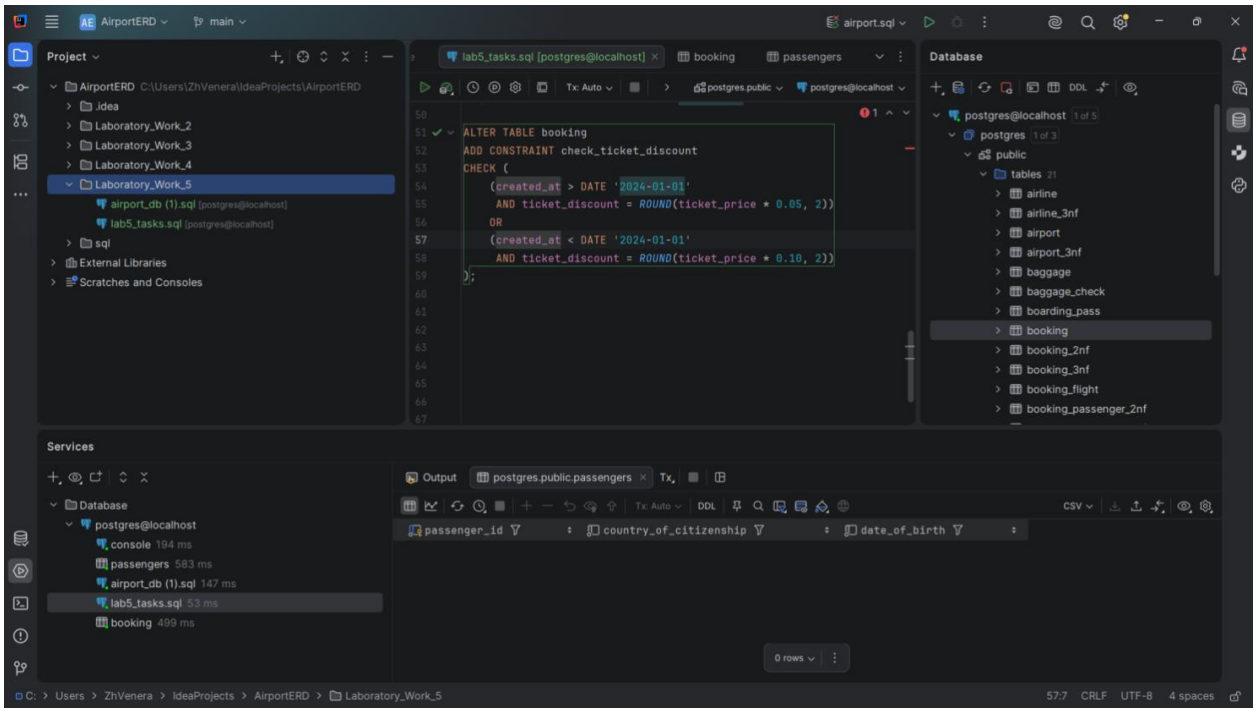
TASK 6


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql [postgres@localhost]
lab5_tasks.sql [postgres@localhost]
sql
External Libraries
Scratches and Consoles
console
lab5_tasks.sql [postgres@localhost]
passengers
Tx: Auto
postgres.public
postgres@localhost
UNIQUE (passport_number);
SELECT passenger_id, first_name, last_name, gender, date_of_birth
FROM passengers
WHERE NOT (
(gender = 'Male' AND date_of_birth <= DATE '2008-10-12')
OR
(gender = 'Female' AND date_of_birth <= DATE '2007-10-12')
);
Database
DDL
postgres@localhost
booking_passenger_2nf
booking_passenger_3nf
booking_receipt
booking_receipt_1nf
flight_3nf
flights
passenger_2nf
passenger_3nf
passengers
security_check
Database Objects
virtual views
Server Objects
Query Files
Services
Output
postgres.public.passengers
Tx
Tx: Auto
DDL
CSV
Database
postgres@localhost
console 194 ms
passengers 583 ms
airport_db (1).sql 147 ms
lab5_tasks.sql 417 ms
assenger_id
first_name
last_name
gender
date_of_birth
1 7 Passenger7 Surname7 Female 2012-02-16
2 8 Passenger8 Surname8 Female 2009-10-19
3 25 Passenger25 Surname25 Male 2012-06-29
4 26 Passenger26 Surname26 Female 2009-07-10
5 34 Passenger34 Surname34 Male 2012-08-06
6 37 Passenger37 Surname37 Male 2011-10-08
32 rows
C: Users ZhVenera IdeaProjects AirportERD Laboratory_Work_5 lab5_tasks.sql
27:3 CRLF UTF-8 4 spaces

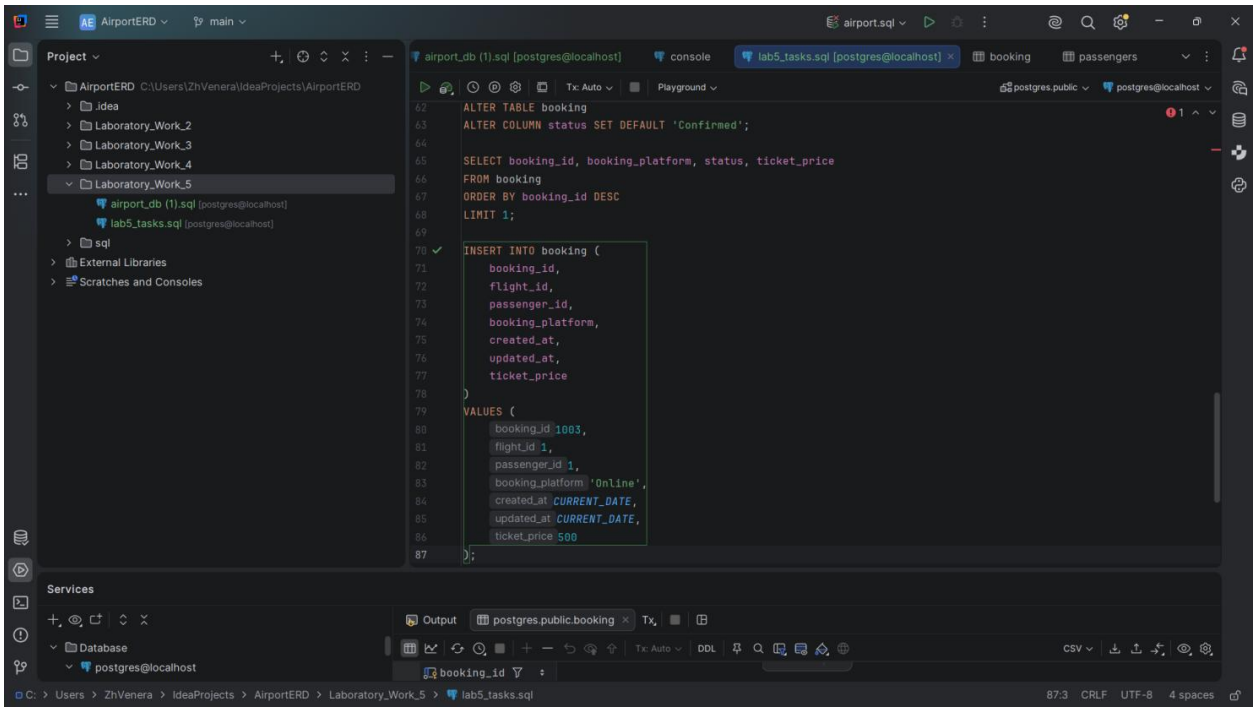
TASK 7

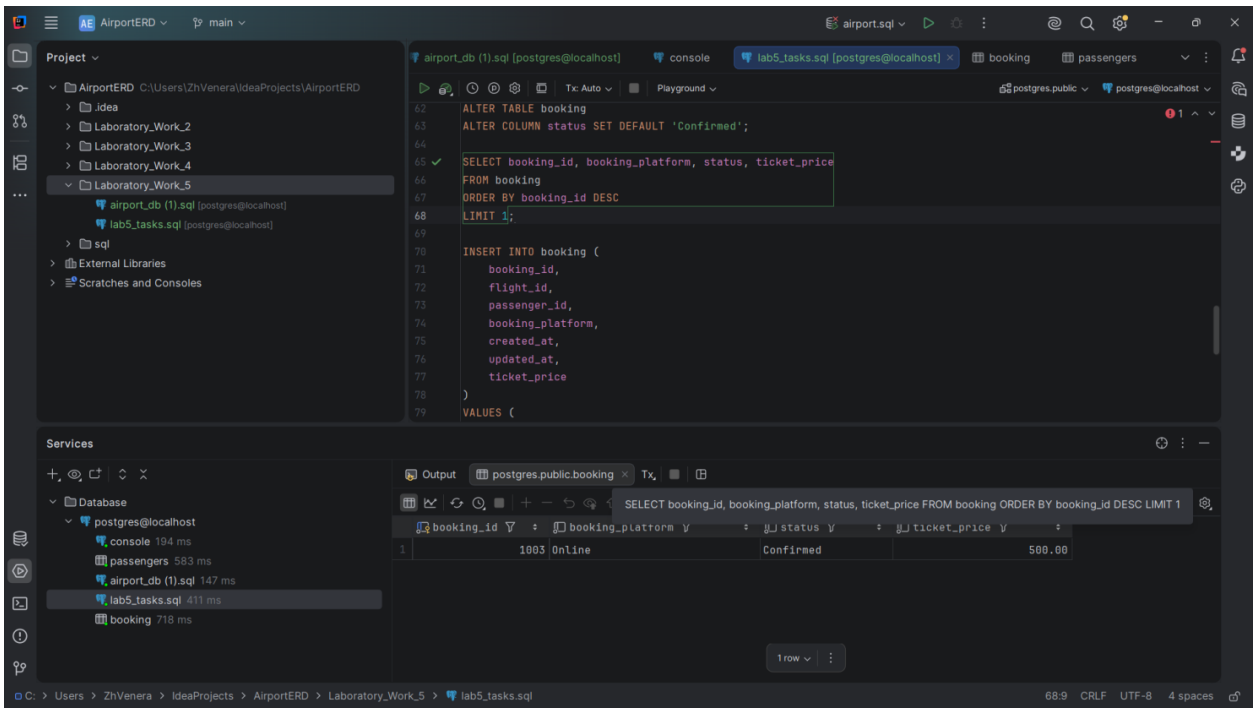

ADD CONSTRAINT check_citizenship_age
AND date_of_birth <= DATE '2007-10-12')
);
SELECT passenger_id, country_of_citizenship, date_of_birth
FROM passengers
WHERE (country_of_citizenship = 'Kazakhstan'
AND date_of_birth > DATE '2008-10-12')
OR (country_of_citizenship = 'France'
AND date_of_birth > DATE '2009-10-12')
OR (country_of_citizenship NOT IN ('Kazakhstan', 'France')
AND date_of_birth > DATE '2007-10-12');

TASK 8

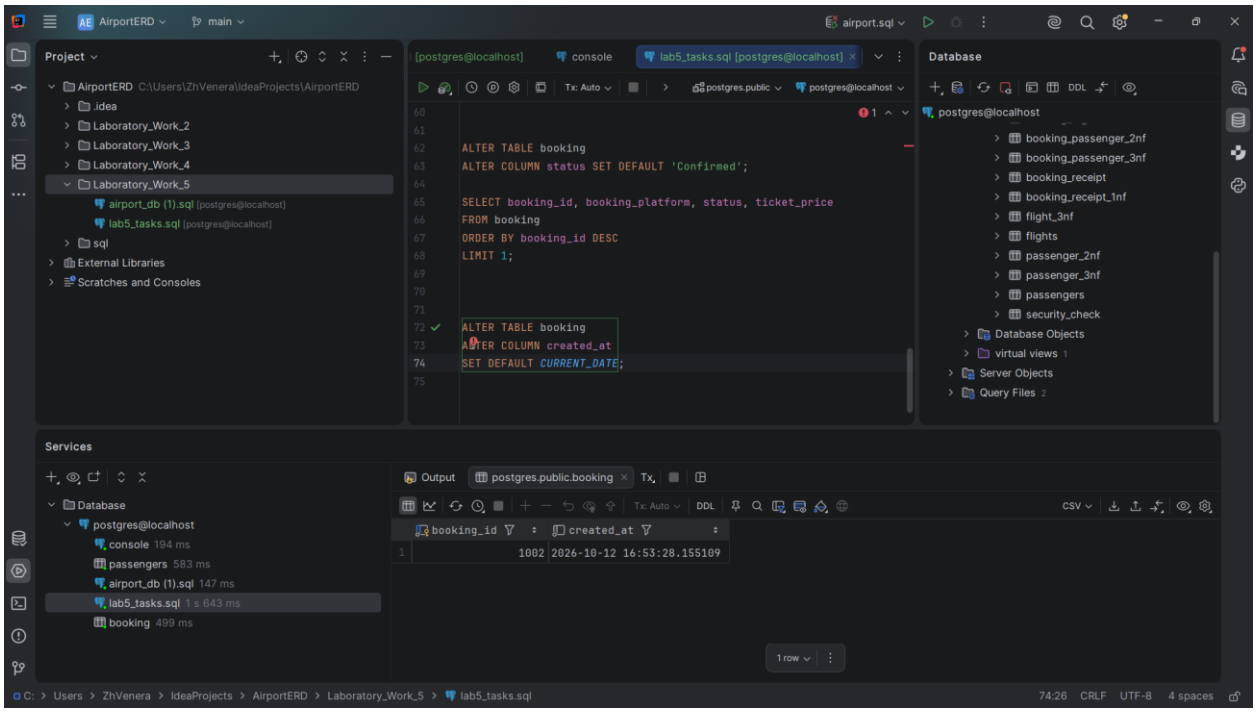

ALTER TABLE booking
ADD CONSTRAINT check_ticket_discount
CHECK (
(created_at > DATE '2024-01-01'
AND ticket_discount = ROUND(ticket_price * 0.05, 2))
OR
(created_at < DATE '2024-01-01'
AND ticket_discount = ROUND(ticket_price * 0.10, 2))
);

TASK 9


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql [postgres@localhost]
lab5_tasks.sql [postgres@localhost]
sql
External Libraries
Scratches and Consoles
airport_db (1).sql [postgres@localhost]
console
lab5_tasks.sql [postgres@localhost]
booking
passengers
Tx: Auto
Playground
postgres.public
postgres@localhost
ALTER TABLE booking
ALTER COLUMN status SET DEFAULT 'Confirmed';
SELECT booking_id, booking_platform, status, ticket_price
FROM booking
ORDER BY booking_id DESC
LIMIT 1;
INSERT INTO booking (
booking_id,
flight_id,
passenger_id,
booking_platform,
created_at,
updated_at,
ticket_price
)
VALUES (
booking_id 1003,
flight_id 1,
passenger_id 1,
booking_platform 'Online',
created_at CURRENT_DATE,
updated_at CURRENT_DATE,
ticket_price 500
);
Services
Output
postgres.public.booking
Database
postgres@localhost
DDL
CSV
booking_id
C: Users ZhVenera IdeaProjects AirportERD Laboratory_Work_5 lab5_tasks.sql
87:3 CRLF UTF-8 4 spaces


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql [postgres@localhost]
lab5_tasks.sql [postgres@localhost]
sql
External Libraries
Scratches and Consoles
airport_db (1).sql [postgres@localhost]
console
lab5_tasks.sql [postgres@localhost]
booking
passengers
ALTER TABLE booking
ALTER COLUMN status SET DEFAULT 'Confirmed';
SELECT booking_id, booking_platform, status, ticket_price
FROM booking
ORDER BY booking_id DESC
LIMIT 1;
INSERT INTO booking (
booking_id,
flight_id,
passenger_id,
booking_platform,
created_at,
updated_at,
ticket_price
)
VALUES (
Services
Database
postgres@localhost
console 194 ms
passengers 583 ms
airport_db (1).sql 147 ms
lab5_tasks.sql 411 ms
booking 718 ms
Output
postgres.public.booking
SELECT booking_id, booking_platform, status, ticket_price FROM booking ORDER BY booking_id DESC LIMIT 1
booking_id
booking_platform
status
ticket_price
1003
Online
Confirmed
500.00
1 row
C: Users ZhVenera IdeaProjects AirportERD Laboratory_Work_5 lab5_tasks.sql
68:9
CRLF
UTF-8
4 spaces

TASK 10


AirportERD
main
airport.sql
Project
AirportERD C:\Users\ZhVenera\IdeaProjects\AirportERD
.idea
Laboratory_Work_2
Laboratory_Work_3
Laboratory_Work_4
Laboratory_Work_5
airport_db (1).sql [postgres@localhost]
lab5_tasks.sql [postgres@localhost]
sql
External Libraries
Scratches and Consoles
[postgres@localhost]
console
lab5_tasks.sql [postgres@localhost]
ALTER TABLE booking
ALTER COLUMN status SET DEFAULT 'Confirmed';
SELECT booking_id, booking_platform, status, ticket_price
FROM booking
ORDER BY booking_id DESC
LIMIT 1;
ALTER TABLE booking
ALTER COLUMN created_at
SET DEFAULT CURRENT_DATE;
Database
postgres@localhost
booking_passenger_2nf
booking_passenger_3nf
booking_receipt
booking_receipt_1nf
flight_3nf
flights
passenger_2nf
passenger_3nf
passengers
security_check
Database Objects
virtual views
Server Objects
Query Files
Services
Database
postgres@localhost
console 194 ms
passengers 583 ms
airport_db (1).sql 147 ms
lab5_tasks.sql 1 s 643 ms
booking 499 ms
Output
postgres.public.booking
booking_id
created_at
1002
2026-10-12 16:53:28.155109
1 row
C: Users ZhVenera IdeaProjects AirportERD Laboratory_Work_5 lab5_tasks.sql
74:26
CRLF
UTF-8
4 spaces